

Java Access Modifiers Practice Questions

Level 1 (Easy)

Access Modifier	Within Class	Within Package	Same Package by subclasses	Outside Package by subclasses	Global
Public	Yes	Yes	Yes	Yes	Yes
Protected	Yes	Yes	Yes	Yes	No
Default	Yes	Yes	Yes	No	No
Private	Yes	No	No	No	No

Question 1: Student Information System

Create a Student class.

Requirements:

- studentName should be public.
- **rollNumber should be private.**
- course should be protected.
- collegeName should have default access.

Create getter and setter methods for rollNumber.

From the main() method:

- Set all values.
- Print all accessible values.
- Access the private variable using getter.

Question 2: Bank Account

Create a class named BankAccount.

Requirements:

- Account Holder Name → public
- Account Number → private
- Balance → private

Methods:

- deposit()
- withdraw()

- checkBalance()

Conditions:

- Balance cannot become negative.
 - Deposit amount cannot be negative.
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Question 3: Employee Salary

Create an Employee class.

Requirements:

- employeeName → public
- employeeId → private
- salary → private

Create methods:

- setSalary()
- getSalary()

Condition:

Salary cannot be less than 15,000.

Question 4: Mobile Phone

Create a Mobile class.

Variables:

- Brand
- Model
- IMEI Number
- Price

Requirement:

Only IMEI Number should be private.

Create getter and setter for IMEI Number.

Question 5: Car Details

Create a Car class.

Variables:

- Brand
- Model
- Engine Number
- Price

Make the Engine Number private.

Display all details using methods.

Level 2 (Medium)

Question 6: ATM Machine

Create an ATM class.

Variables:

- accountHolder
- pin
- balance

Requirements:

- PIN must be private.
- Balance must also be private.

Methods:

- login()
- withdraw()
- deposit()
- checkBalance()

Condition:

Withdrawal should only happen after entering the correct PIN.

Question 7: Hospital Management

Create a Patient class.

Variables:

- patientName

- patientId
- medicalHistory
- doctorName

Requirements:

Medical History should be private.

Create methods:

- addMedicalHistory()
 - displayPatient()
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Question 8: College Result System

Create a StudentResult class.

Variables:

- Student Name
- Roll Number
- Marks

Requirements:

Marks should be private.

Methods:

- setMarks()
 - calculateGrade()
 - displayResult()
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Question 9: E-Commerce Product

Create a Product class.

Variables:

- Product Name
- Product ID

- Price
- Stock

Requirements:

Price and Stock should be private.

Methods:

- purchase()
- addStock()
- displayProduct()

Conditions:

Stock should never become negative.

Question 10: Movie Ticket Booking

Create a MovieTicket class.

Variables:

- Movie Name
- Seat Number
- Ticket Price
- Booking Status

Requirements:

Booking Status should be private.

Methods:

- bookTicket()
 - cancelTicket()
 - showStatus()
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Level 3 (Advanced)

Question 11: Online Banking System

Create two classes.

Account

Private Variables:

- accountNumber

- balance
- PIN

Public Methods:

- deposit()
 - withdraw()
 - changePIN()
 - checkBalance()
-

Bank

Methods:

- transferMoney()
 - createAccount()
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Question 12: Employee Payroll

Create classes:

Employee

Private Variables:

- Salary
- Bonus

Protected Variables:

- Department

Public Variables:

- Name

Create Payroll class to calculate monthly salary.

Question 13: Library Management

Classes:

- Book
- Student
- Library

Requirements:

Book ID should be private.

Methods:

- issueBook()
 - returnBook()
 - searchBook()
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Question 14: School Management System

Classes:

- Student
- Teacher
- Principal

Requirements:

Fees should be private.

Only the Principal can update fees.

Students can only view fees.

Question 15: Online Shopping Cart

Create classes:

- Product
- Customer
- Cart

Requirements:

Product Price should be private.

Methods:

- addProduct()
 - removeProduct()
 - calculateBill()
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Level 4 (Package-Based Access Modifier Practice)

These questions help students understand **public**, **private**, **protected**, and **default** across packages.

Question 16: University Management

Package 1:

- Student

Package 2:

- Main

Try accessing:

- public
- protected
- default
- private variables

Observe which variables are accessible.

Question 17: Animal Management

Package:

animals

Classes:

- Animal
- Dog

Another Package:

zoo

Class:

- Zoo

Use different access modifiers and verify accessibility.